

Simulation-Based Ant Colony Optimization for Complex System Configuration Problems

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Abstract—Complex system configuration problems are the problems of appropriately assigning system parameter values for optimizing some aspect of complex system performance. In this paper, we first cast complex system configuration problems as mixed-variable parameter optimization problems where mensurable system simulation responses are used for evaluation. Then we present a simulation-based ant colony optimization algorithm ($sACO_{MV}$) to tackle the problems. In $sACO_{MV}$, the decision variables of the complex system configuration problems can be clearly declared as continuous, ordinal, or categorical and let the algorithm treat them adequately. Finally, $sACO_{MV}$ is tested on mixed-variable complex engineering system configuration problems. The effectiveness and robustness of $sACO_{MV}$ are demonstrated by the comparisons with results from the literature.

Keywords—Ant colony optimization; complex system configuration problems; mixed variables; simulation responses

I. INTRODUCTION

A complex system often has parameters whose settings play an important role on its performance and behavior. An instantiation of such a system that requires specific values assigned to all parameters is referred to as a *system configuration*. System configuration problems are the problems of selecting the optimal system configurations. In many cases, practitioners manually and iteratively configure systems by running simulations of different parameter settings. This configuration process is usually based on personal experience and time-consuming. Automating this process is therefore of highly practical relevance.

In this paper, we first cast complex system configuration problems as automatic parameter optimization tasks, for which highly performing optimization techniques can be applied to determine the optimal parameter settings given the mensurable system simulation responses. In terms of the paradigm of optimization problems, we model the complex system configuration problems using combinations of continuous and discrete variables. The discrete variables in these problems can be ordinal or categorical. Continuous variables take values from a subset of the real numbers. Ordinal variables exhibit a natural ordering relation (e.g., integers of system components). Categorical variables take values from a finite set of unordered categories (e.g., shapes or types of system topologies).

We then present a simulation-based ant colony optimization algorithm ($sACO_{MV}$) to automatically tackle the mixed-variable complex system configuration problems. $sACO_{MV}$,

allows to explicitly declare each decision variable of a complex system configuration problem as continuous, ordinal or categorical and use mensurable system simulation responses to evaluate system configurations. $sACO_{MV}$ includes three solution construction approaches: a continuous optimization approach ($ACO_{\mathbb{R}}$) for continuous variables, a continuous relaxation approach (ACO_{MV-o}) for ordinal variables, and a categorical optimization approach (ACO_{MV-c}) for categorical variables. Together, these approaches allow $sACO_{MV}$ to tackle mixed-variable problems. As a final step, we apply $sACO_{MV}$ to complex mixed-variable engineering system configuration problems. The experimental results demonstrate the effectiveness and robustness of $sACO_{MV}$.

The paper is organized as follows. Section II formulates mixed-variable system configuration problems. Section III introduces the ant colony optimization metaheuristic and Section IV describes $sACO_{MV}$. In Section V, we apply $sACO_{MV}$ to engineering system configuration problems and compare the results obtained with $sACO_{MV}$ to those obtained by other algorithms. In Section VI we conclude and give directions for future work.

II. MIXED-VARIABLE COMPLEX SYSTEM CONFIGURATION PROBLEMS

Complex system configuration problems consist in how to find the appropriate system parameters that result in the best system performance. We cast complex system configuration problems as automatic parameter optimization problems. Continuous system configuration problems are system parameter optimization problems where all variables have a domain that typically is a subset of the real numbers; mixed-variable system configuration problems have additionally other types of variables, so that some variables are continuous and others can be ordinal or categorical.

A model for a mixed-variable system configuration problem (MVSCP) may be defined as follows:

Definition A model $R = (S, \Omega, f)$ of a MVSCP consists of

- a system configuration search space S defined over a finite set of both discrete and continuous variables and a set Ω of constraints among the variables;
- A mensurable system simulation response (objective function) $f : S \rightarrow \mathbb{R}_0^+$ to be minimized.

The system configuration search space $\mathbf{S}$ is defined by a set of $n = d + r$ variables $x_i, i = 1, \dots, n$, of which d are discrete and r are continuous. The discrete variables include o ordinal variables and c categorical ones, $d = o + c$. A system configuration solution $S \in \mathbf{S}$ is a complete value assignment, that is, each decision variable is assigned a value. A feasible solution is a solution that satisfies all constraints in the set Ω . A global optimum $S^* \in \mathbf{S}$ is a feasible solution that satisfies $f(S^*) \leq f(S) \forall S \in \mathbf{S}$. The set of all globally optimal solutions is denoted by $\mathbf{S}^*, \mathbf{S}^* \subseteq \mathbf{S}$. Solving an MVSCP requires finding at least one $S^* \in \mathbf{S}^*$.

Given measurable system simulation responses, we give a flowchart of using optimization technique for solving mixed-variable complex system configuration problems in Fig. 1.

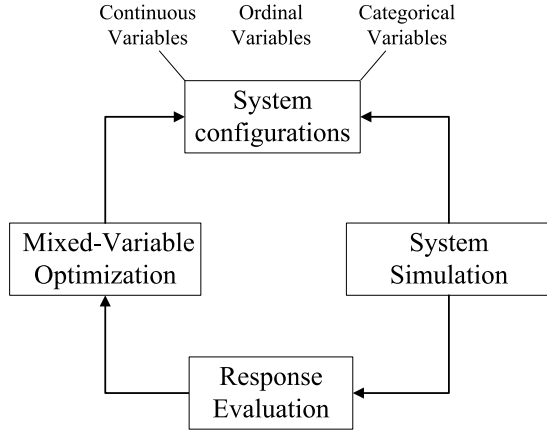


Fig. 1. A flowchart of using optimization technique for solving mixed-variable complex system configuration problems given the measurable system simulation responses

III. ANT COLONY OPTIMIZATION METAHEURISTIC

Ant Colony Optimization (ACO) algorithms were first proposed for tackling discrete optimization problems [1], [2]. ACO algorithms for discrete optimization problems make use of a so-called pheromone model in order to probabilistically construct solutions. A pheromone model consists of a set of numerical values, called pheromones, that are a function of the search experience of the algorithm. The pheromone model is used to bias the solution construction towards regions of the search space containing high quality solutions.

The ACO metaheuristic [3], [4] defines a class of optimization algorithms inspired by the foraging behavior of real ants. The main algorithmic components of the ACO metaheuristic are the ants' solution construction and the update of the pheromone information. The ACO metaheuristic finds approximate solutions to an optimization problem mainly by iterating the following two steps: 1) candidate solutions are probabilistically constructed using a probability distribution over the search space; 2) good candidate solutions are used to modify the probability distribution so that future sampling is biased toward higher quality solutions. After the initial proposals of ACO algorithms for combinatorial problems, its adaptation to solve continuous or integer optimization

problems enjoys an increasing attention [5], [6], [7], [8], [9]. The first algorithm that can be classified as an ACO algorithm for continuous domains is $\text{ACO}_{\mathbb{R}}$ [5].

IV. sACO_{MV} FOR MIXED-VARIABLE SYSTEM CONFIGURATION PROBLEMS

This section introduces the algorithm structure of sACO_{MV} and describes the ant probabilistic solution construction for continuous variables, ordinal variables and categorical variables, respectively.

A. sACO_{MV} structure

In ACO for discrete optimization problems, the pheromone values associate with a finite set of discrete components. This is not possible when continuous or mixed continuous-discrete variables are involved. Therefore, sACO_{MV} makes use of a *solution archive*, called *SA*, as a form of pheromone model by following the principle of population-based ACO [10]. *SA* keeps track of complete solutions of the system configuration problem.

Given a mixed variable system configuration problem of n -dimensions and k system configuration solutions, sACO_{MV} stores the values of the n variables and the measurable system simulation response (objective function) value of each k solution in the *SA*. Fig. 2 shows the structure of the *SA*, which includes three groups of columns, one group for continuous variables, one group for ordinal variables and another one for categorical variables.

The description of the sACO_{MV} algorithm is given as follows. The *SA* is initialized with k random solutions. Then, these k solutions are kept sorted according to their quality (from best to worst). Solution S_j has associated a weight ω_j . This weight is computed using a Gaussian function as:

$$\omega_j = \frac{1}{\alpha k \sqrt{2\pi}} e^{-\frac{(\text{rank}(j)-1)^2}{2\alpha^2 k^2}}, \quad (1)$$

where $\text{rank}(j)$ is the rank of solution S_j in the sorted archive, and α is a parameter of the algorithm. By calculating $\text{rank}(j) - 1$, which corresponds to setting the mean of the Gaussian function to 1, the best solution receives the highest weight, while the weight of the other solutions decreases in an exponential way with their rank. At each iteration of the algorithm, m ants probabilistically construct m new solutions, where an ant is a probabilistic solution construction procedure. Once the m solutions have been constructed, they are added into the *SA*. The $k + m$ solutions in the *SA* are then sorted and the m worst ones are removed. So, a new solution archive is constituted by the remaining k solutions. Accordingly, the algorithm search bias towards the best solutions found during the search process. During the ant probabilistic solution construction process, an ant uses construction approach $\text{ACO}_{\mathbb{R}}$ to deal with continuous variables, $\text{ACO}_{\text{MV-o}}$ and $\text{ACO}_{\text{MV-c}}$ to deal with ordinal variables and categorical variables, respectively. Their details are described in the following subsection. Algorithm 1 shows an outline of the sACO_{MV} algorithm. The functions *Best* and *Sort* in Algorithm 1 mean the sorting of the archive and the selection of the k best solutions. sACO_{MV} does not consider the design of restart strategy in this paper.

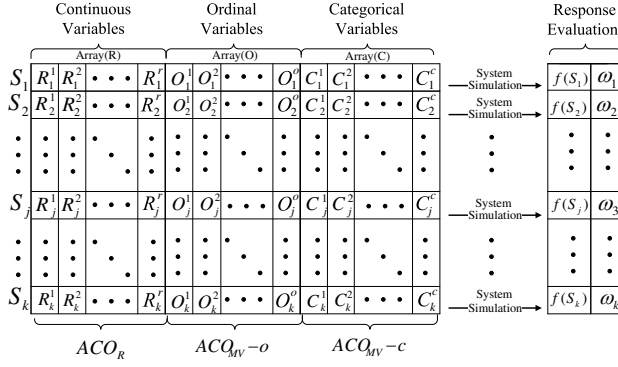


Fig. 2. The structure of SA used by sACOMV for complex system configuration problems. The measurable system simulation responses (objective function) are used to evaluate system configuration solutions.

Algorithm 1 sACOMV outline

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Initialize  $k$  solutions
Evaluate them by system simulation responses
{Sort solutions and store them in SA}
SA  $\leftarrow$  Sort( $S_1 \dots S_k$ )
while termination criterion is not satisfied do
  for 1 to  $m$  do
    Ant probabilistic solution construction for  $ACO_R$ 
    Ant probabilistic solution construction for  $ACO_{MV-o}$ 
    Ant probabilistic solution construction for  $ACO_{MV-c}$ 
    Store newly constructed solution
    Evaluate it by system simulation responses
  end for
  {Sort solutions and select the best  $k$  solutions}
  SA  $\leftarrow$  Best(Sort( $S_1 \dots S_{k+m}$ ),  $k$ )
end while

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B. Probabilistic Solution Construction for Continuous Variables

Continuous variables are handled by ACO_R [5]. In ACO_R , the construction of new solutions by the ants is incrementally accomplished, variable by variable. First, an ant probabilistically chooses one solution from SA. The probability of choosing solution j is defined by:

$$p_j = \frac{\omega_j}{\sum_{l=1}^k \omega_l}, \quad (2)$$

where ω_j is computed by Equation (1).

An ant then constructs a new continuous variable solution around the chosen solution j . At the i -th construction step, $1 \leq i \leq r$, an ant assigns a value to continuous variable i by sampling the neighborhood around the value R_j^i of the chosen j -th solution. The sampling is done using a Gaussian probability density function with mean μ and standard deviation σ :

$$g(x, \mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}. \quad (3)$$

Set $\mu = R_j^i$ when considering continuous variable i of solution j . Furthermore, set

$$\sigma = \beta \sum_{l=1}^k \frac{|R_l^i - R_j^i|}{k-1}, \quad (4)$$

which is the average distance between the values of the i -th continuous variable of the solution j and the values of the i -th continuous variables of the other solutions in the SA, multiplied by a parameter β . This parameter has a similar effect to the pheromone persistence parameter in ACO algorithms. The lower the value of β , the higher the convergence speed of the algorithm. This process repeats by each dimension for each of the m ants. sACOMV also considers a non-deterministic adaptive method [5] to deal with the correlation between the decision variables.

C. Probabilistic Solution Construction for Ordinal Variables

If the complex system configuration problem being considered includes ordinal variables, the continuous relaxation approach, ACO_{MV-o} , is used. ACO_{MV-o} do operate the indices in an array of the actual values of the ordinal variables. The values of the indices for the new solutions are constructed as real numbers, as it is the case for the continuous variables. But, the continuous values are rounded to the nearest valid index before the system simulation response (objective function) is evaluated, and the value at that index is then used for the evaluation. The reason for this choice is that ordinal variables do not necessarily have numerical values like integers; for example, an ordered variable may take as possible values {small, medium, large}. Otherwise, ACO_{MV-o} works exactly as what ACO_R does.

D. Probabilistic Solution Construction for Categorical Variables

While ordinal variables are relaxed and handle by ACO_R , categorical variables are handled differently by ACO_{MV-c} [9] as this type of variables has no predefined ordering. At each step of ACO_{MV-c} , an ant assigns a value to one variable at a time. For each categorical variable i , $1 \leq i \leq c$, an ant probabilistically chooses one of the t_i available values $v_l^i \in \{v_1^i, \dots, v_{t_i}^i\}$. The probability of choosing the l -th value is defined by

$$p_l^i = \frac{H_l}{\sum_{j=1}^{t_i} H_j}, \quad (5)$$

where H_l is the weight that associates with the l -th available value. The weight H_l is computed as

$$H_l = \begin{cases} \frac{\omega_{j_l}}{u_l^i} + \frac{\alpha}{\eta}, & \text{if } (\eta > 0, u_l^i > 0), \\ \frac{\omega_{j_l}}{u_l^i}, & \text{if } (\eta = 0, u_l^i > 0), \\ \frac{\alpha}{\eta}, & \text{if } (\eta > 0, u_l^i = 0), \end{cases} \quad (6)$$

where ω_{j_l} is computed according to Equation (1) with j_l being the index of the highest quality solution that uses value v_l^i for the categorical variable i . u_l^i is the number of solutions that use value v_l^i for the categorical variable i in the SA (hence, the more common the value v_l^i is, the lower is its

final weight); thus, u_l^i is a variable whose value is adapted during run-time and that controls the weight of choosing the l -th available value. $u_l^i = 0$ corresponds to the case in which the l -th available value is not used by the solutions in the SA; in this case the weight of the l -th value is $\frac{\alpha}{\eta}$. η is the number of values from the t_i available ones that are not used by the solutions in the SA; $\eta = 0$ (that is, all values are used) corresponds to the case in which $\frac{\alpha}{\eta}$ is discarded. Again, η is a variable that is adapted during run-time and, if $\eta = 0$, it is natural to discard the second component in Equation (6). Please notice that u_l^i and η are nonnegative numbers, and their values are never equal to zero at the same time. α is the same parameter of the algorithm that was used in Equation (1).

V. APPLICATION TO MIXED-VARIABLE ENGINEERING SYSTEM CONFIGURATION PROBLEMS

sACOMV is tested on mixed-variable complex engineering system configuration problem that deals with continues, ordinal or categorical variables. For our experiments, the parameter settings from Table I was used for sACOMV. The parameter settings were automatically tuned by Iterated F-Race [11] on a set of artificial, mixed-variable benchmark functions [9], [12], [13]. The most fundamental constraint handling technique was used, which consists in rejecting all infeasible solutions in the optimization process (also called “death penalty”). 100 independent runs were performed for each engineering system configuration problem.

TABLE I. PARAMETER SETTINGS OF sACOMV USED FOR TESTING ON MIXED-VARIABLE COMPLEX ENGINEERING SYSTEM CONFIGURATION PROBLEMS

Number of ants	Influence of best solutions	Search width	SA size
m	α	β	k
5	0.05099	0.6795	90

A. Pressure vessel system configuration problem

The pressure vessel system requires configurations of a cylindrical body and two hemispherical heads to minimize the manufacturing cost. The schematic picture of the vessel is presented in Fig. 3. Two continues variables and two ordinal variables are involved in the simulation models. They are the thickness of the main cylinder T_s , the thickness of the heads T_h , the inner radius of the main cylinder R , and the length of the main cylinder L . Variables R and L are continuous and the thickness for variables T_s and T_h are chosen only from a set of allowed values, these being the integer multiples of 0.0625 inch. The simulation models of the three cases A, B, C is given in Table II. These cases differ by the constraints posed on the thickness of the steel used for the heads and the main cylinder.

In case A, B and C (see Table IV), sACOMV obtains the best-known solution value with a success rate of 100% using maximum of 1737, 1764 and 1666 simulation evaluations, respectively, while other algorithms do not reach a success rate of 100% with respect to the best-known solution value even after many more number of evaluations. The run-time behavior of sACOMV by using run-length distributions (RLDs, for short) [14] is shown in Fig. 5. The (empirical) RLD provides a

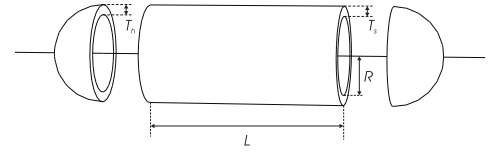


Fig. 3. Schematic view of the pressure vessel system.

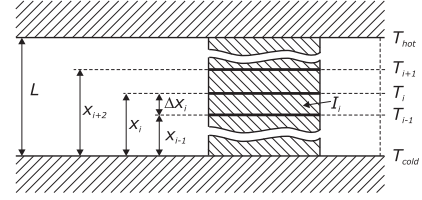


Fig. 4. Schematic view of the thermal insulation system.

graphical view of the development of the empirical frequency of finding a solution of a certain quality as a function of the number of simulation evaluations.

B. Thermal insulation system configuration problem

A thermal insulation system requires configurations of the locations and temperatures of the intercepts, the number of intercepts, and the material types of insulators placed between each pair of neighboring intercepts. Fig. 4 shows the schema of the thermal insulation system. In this system, the heat flow is minimized from a hot to a cold surface by heat intercepts. The heat is intercepted by imposing a cooling temperature T_i at locations x_i , $i = 1, 2, \dots, n$. The simulation model of the classic model of thermal insulation systems is given in Table III. The effective thermal conductivity k of all these insulators changes with the temperature and does so differently for different materials. Given that the number of intercepts n is set based on the model presented ($n=10$), the problem variables may be defined as: $I_i \in \mathbf{M}$, $i = 1, \dots, n + 1$ — the material used for the insulation between the $(i - 1)$ -th and the i -th intercepts (from a set of $\mathbf{M}$ materials). $\Delta x_i \in \mathbb{R}_+$, $i = 1, \dots, n + 1$ — the thickness of the insulation between the $(i - 1)$ -th and the i -th intercepts. $\Delta T_i \in \mathbb{R}_+$, $i = 1, \dots, n + 1$ — the temperature difference of the insulation between the $(i - 1)$ -th and the i -th intercepts. Accordingly, there are $n + 1$ (=11) categorical variables and the other $2n + 2$ (=22) continuous variables.

In the experimental results (see Table V), sACOMV obtains a better solution than MVP [15] and FMGPS [16] and a new best-known simulation response evaluation value, 24.148. Fig.6 also shows the evolution of the best solution as a function of number of simulation evaluations of sACOMV for details.

VI. CONCLUSIONS AND FUTURE WORK

In this paper, we cast complex system configuration problems as automatic mixed-variable parameter optimization problems for which measurable system simulation responses are used for evaluating system configurations. We introduce sACOMV, a simulation-based ant colony optimization algorithm (sACOMV) for automatically tackling the mixed-variable parameter optimization problems. The experimental results for the mixed variable complex engineering system

TABLE II. THE SIMULATION MODELS OF THE CASES (A, B, C) OF THE PRESSURE VESSEL SYSTEM CONFIGURATION PROBLEM.

No	Case A	Case B	Case C
	$\min f = 0.6224 T_s R L + 1.7781 T_h R^2 + 3.1611 T_s^2 L + 19.84 T_s^2 R$		
g_1		$-T_s + 0.0193 R \leq 0$	
g_2		$-T_h + 0.00954 R \leq 0$	
g_3		$-\pi R^2 L - \frac{4}{3}\pi R^3 + 750 \cdot 1728 \leq 0$	
g_4		$L - 240 \leq 0$	
g_5	$1.1 \leq T_s \leq 12.5$	$1.125 \leq T_s \leq 12.5$	$1 \leq T_s \leq 12.5$
g_6	$0.6 \leq T_h \leq 12.5$		$0.625 \leq T_h \leq 12.5$
g_7		$0.0 \leq R \leq 240$	
g_8		$0.0 \leq L \leq 240$	

TABLE III. THE SIMULATION MODEL FOR THE THERMAL INSULATION SYSTEM CONFIGURATION PROBLEM.

$f(\mathbf{x}, \mathbf{T}) = \sum_{i=1}^n P_i$	
$= \sum_{i=1}^n AC_i \left(\frac{T_{\text{hot}}}{T_i} - 1 \right) \left(\frac{f_{T_i+1}^{kdT}}{\Delta x_i} - \frac{f_{T_i-1}^{kdT}}{\Delta x_{i-1}} \right)$	
	Constraint
g_1	$\Delta x_i \geq 0, i = 1, \dots, n+1$
g_2	$T_{\text{cold}} \leq T_1 \leq T_2 \leq \dots \leq T_{n-1} \leq T_n \leq T_{\text{hot}}$
g_3	$\sum_{i=1}^{n+1} \Delta x_i = L$
where	$C = 2.5$ if $T \geq 71$ K
	$C = 4$ if $71 \text{ K} > T > 4.2$ K
	$C = 5$ if $T \leq 4.2$ K

TABLE IV. RESULTS FOR CASE A, B AND C OF THE PRESSURE VESSEL SYSTEM CONFIGURATION PROBLEM. f_{Best} DENOTES THE BEST OBJECTIVE FUNCTION VALUE. SR_B DENOTES THE SUCCESS RATE OF REACHING THE BEST KNOWN SOLUTION VALUE. FEs DENOTES THE MAXIMUM NUMBER OF OBJECTIVE FUNCTION EVALUATIONS IN EACH ALGORITHM RUN. REPORT $sACO_{MV}$ IN PARENTHESIS THE LARGEST NUMBER OF OBJECTIVE FUNCTION EVALUATIONS IT REQUIRED IN ANY OF THE 100 INDEPENDENT RUNS ($sACO_{MV}$ REACHED IN EACH RUN THE BEST KNOWN SOLUTION VALUE). “-” MEANS THAT THE INFORMATION IS NOT AVAILABLE.

Case A	f_{Best}	SR_B	FEs
NLIDP [17]	7867.0	-	-
MIDCP [18]	7790.588	-	-
DE [19]	7019.031	89.2%	10000
$sACO_{MV}$	7019.031	100%	(1737)
Case B	f_{Best}	SR_B	FEs
NLIDP [17]	7982.5	-	-
SLA [20]	7197.734	-	-
GA [21]	7207.497	-	-
DE [19]	7197.729	90.2%	10000
HSIA [22]	7197.9	-	-
$sACO_{MV}$	7197.729	100%	(1764)
Case C	f_{Best}	SR_B	FEs
NLMDP [23]	7127.3	-	-
EP [24]	7108.616	-	-
ES [25]	7006.9	-	4800
DE [19]	7006.358	98.3%	10000
CHOPA [26]	7006.51	-	10000
$sACO_{MV}$	7006.358	100%	(1666)

TABLE V. COMPARISON OF THE BEST SIMULATION RESPONSE VALUE FOR THE THERMAL INSULATION SYSTEMS CONFIGURATION PROBLEM.

Objective function	MVP [15]	FMGPS [16]	$sACO_{MV}$
Power($\frac{PL}{A} (\frac{W}{cm})$)	25.294	25.58	24.148

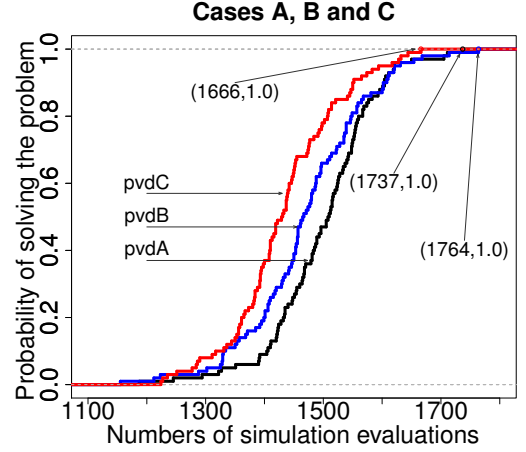


Fig. 5. The RLDs of $sACO_{MV}$ for the pressure vessel design problem case A, B and C (pvdA, pvdB and pvdC are the abbreviations of those problems, respectively).

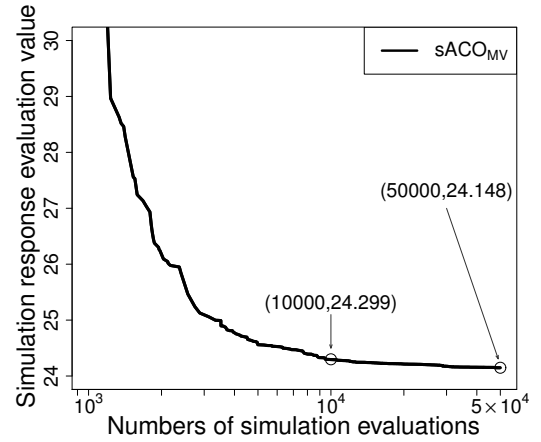


Fig. 6. The development of the best solution quality over the number of simulation evaluations for $sACO_{MV}$ on the thermal insulation systems configuration problem.

configuration problems shows that $sACO_{MV}$ can tackle various decision variables robustly, but also that it is efficient and effective. In the future work, we would place further emphasis on expensive and black-box system-of-systems [27]) configuration problems where the measurement of system simulation is computationally expensive to evaluate, and simulation model is a black-box model. To do so, we intend to research on surrogate model assisted $sACO_{MV}$ using Kriging approximation techniques [28].

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